B.Sc. (Hons) in IT Specialization Interactive Media 2024

Year 04 – Semester 02

SE4031 – GAME DEVELOPMENT

Assignment 02

2D game – Broken Worrier

Submitted By:

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Broken Warrior

Game Documentation

1. Summary of the Game Concept

Broken Warrior: 2D action game featuring intense sword combat. The player controls a legendary warrior charging forward through time, fending off endless waves of corrupted samurai enemies. The game demands sharp reflexes as players chain together 3-hit weapon combos, collect procedurally animated health packs to survive, and strive to reach a score of 100 to achieve ultimate victory.

2. Description of How Core Features are Implemented

1.Modular Character Controller & Physics:

The player is controlled via a custom PlayerMovement.cs script using Unity's Rigidbody2D for physics-based movement and jumping. Ground detection is handled using Physics2D.OverlapCircle to prevent infinite jumping.

2. Endless Enemy Spawning System:

The game features an endless challenge mode powered by EnemySpawner.cs and HealthPackSpawner.cs scripts. These scripts spawn enemies and health packs ahead of the player to create a continuous gameplay loop.

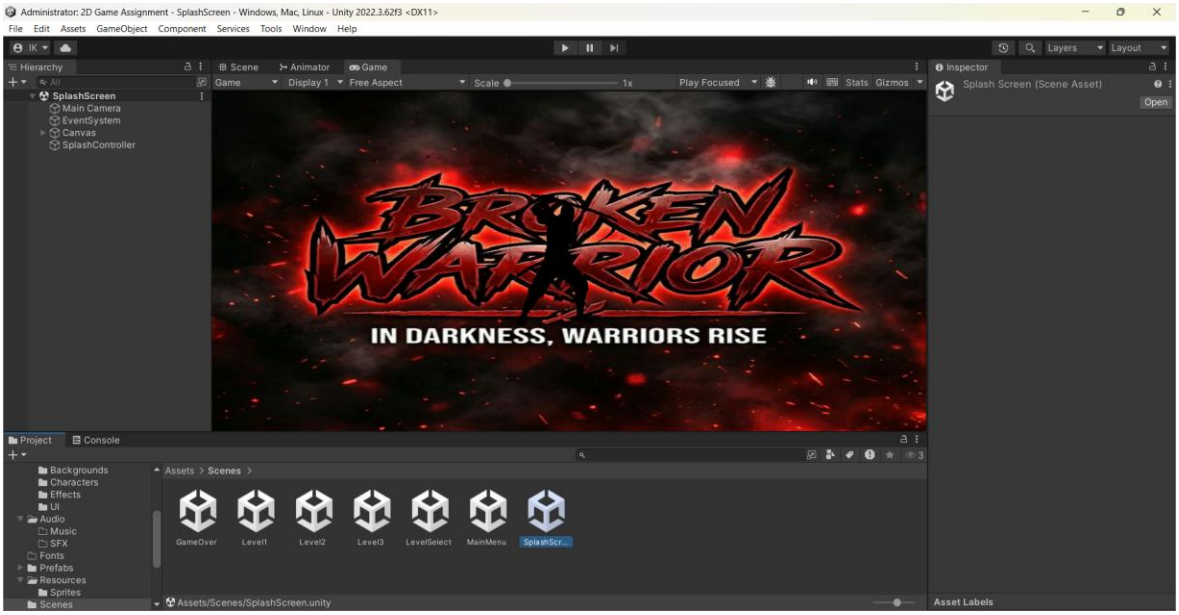
3. State-Machine Combat & Hit Detection:

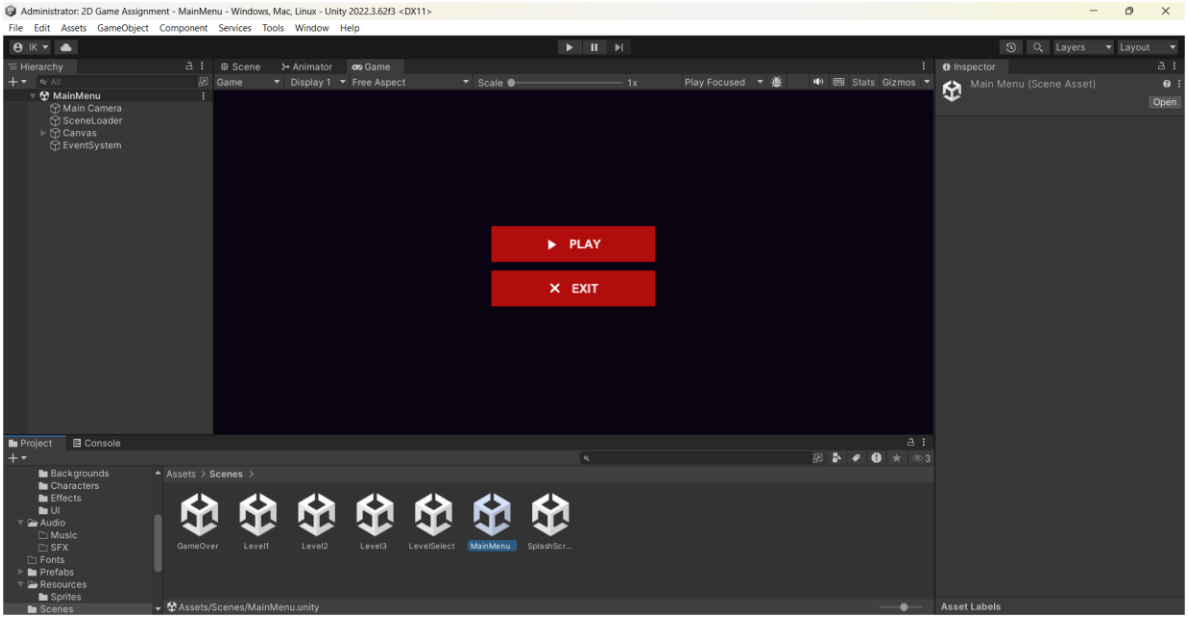
The combat system uses a State Machine in PlayerCombat.cs allowing multiple attack types such as Slash, Chop, Stab, Upper, Low, and Power. Damage detection is handled using Physics2D.OverlapCircle.

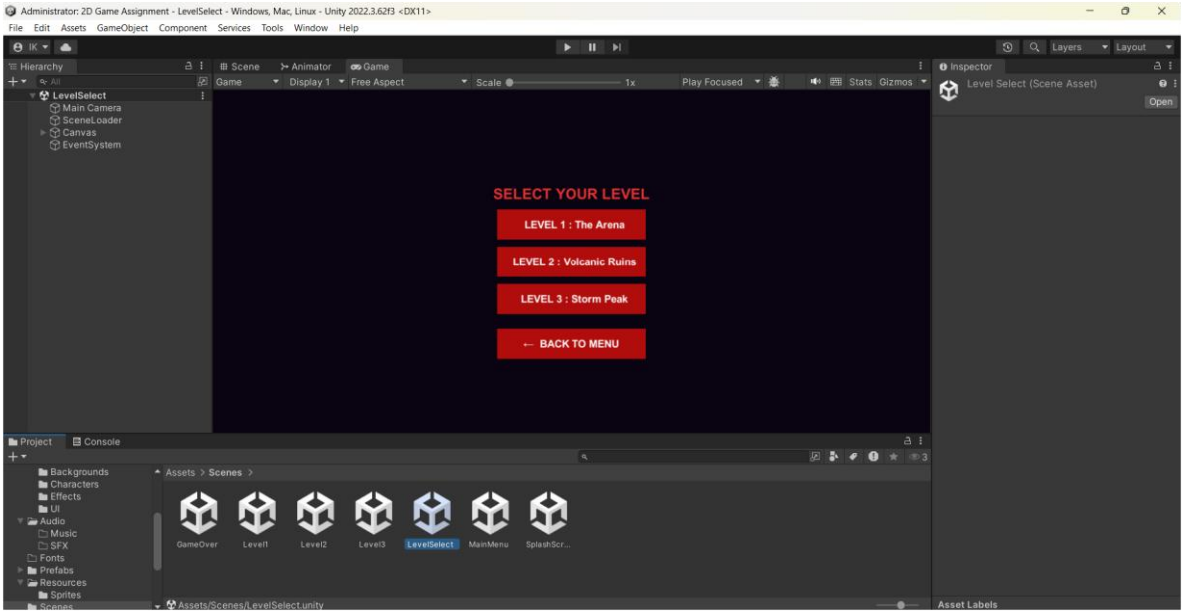
4. Automated Game Flow & Scene Management:

A GameManager system tracks player score and health. When health reaches 0 or score reaches 100, the game transitions to Game Over or Victory scene.

3. Screenshots of Gameplay



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LEVEL 2 : Volcanic Ruins
LEVEL 3 : Storm Peak
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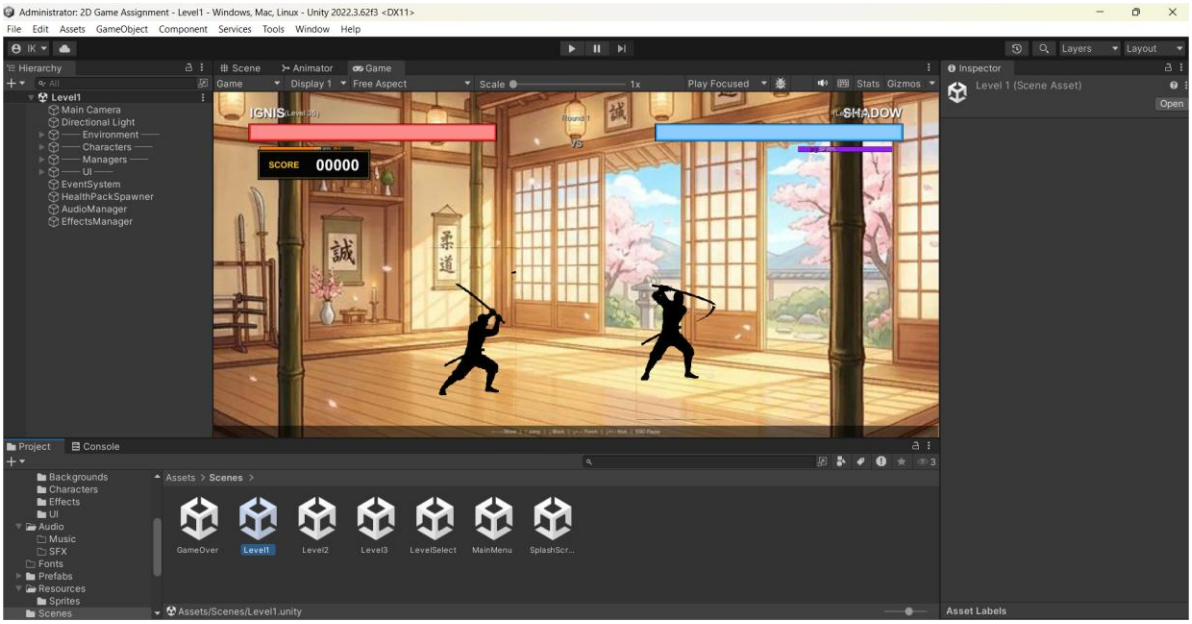
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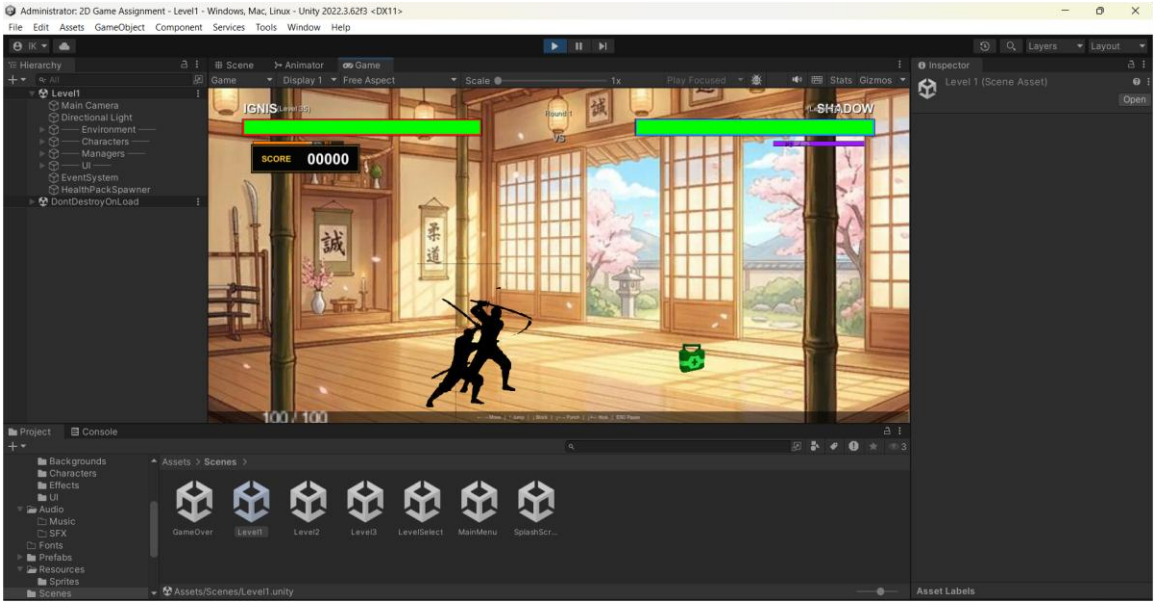

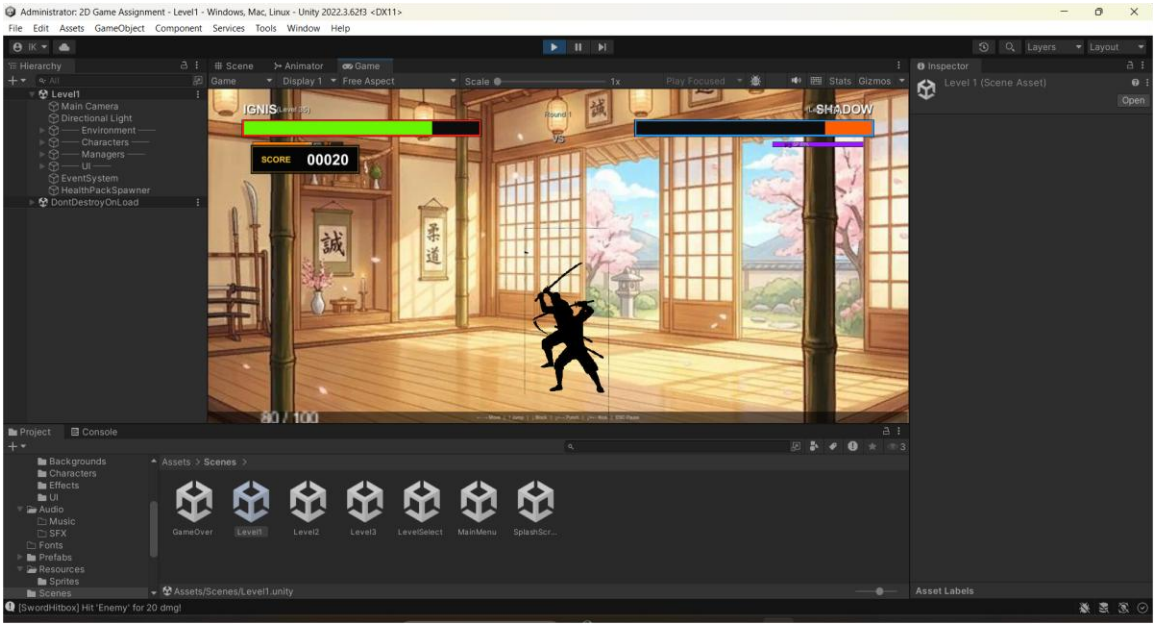

4. Control Guide

- Move Right / Run: D or Right Arrow
- Move Left: A or Left Arrow
- Jump: Spacebar
- Light Attack: Left Mouse Click
- Heavy Attack: Right Mouse Click

- Block / Defend: Shift
- UI Navigation: Mouse Cursor

5. Description of the Creative Feature

1. Procedural Shadow-Fight Style Weapon Animations:
2. The game uses procedural animation via C# Coroutines instead of traditional sprite animations.
3. The weapon animation is generated using mathematical interpolation to create Windup, Strike, and Recover phases. This reduces memory usage and creates smooth and responsive combat effects with slash trail effects.

6. Credits for External Assets/Tools Used

- Game Engine: Unity Editor 2022.3
- Physics Engine: Unity Box2D
- Sprites & Art: Unity Assets / itch.io assets
- Fonts: Arial
- Audio: Custom / Free sound assets

7. Video Link

Drive: GD - IT22606556